Unit (IV) : Multiple Integral [C&V]

① Evaluate $\int_0^3 \int_0^1 (x^2+3y^2)\, dy\, dx$ **Ans : 12**

② Evaluate $\int_0^{\pi/4} \int_0^{\sqrt{\cos 2x}} \frac{y}{(1+y^2)^2}\, dy\, dx$ **Ans : $\frac{\pi-2}{8}$**

③ Evaluate $\int_0^1 \int_0^y xy\, e^{x^2}\, dx\, dy$ **Ans $\frac{1}{4e}$**

④ Evaluate $\iint_R e^{ax+by}\, dy\, dx$ **Ans : $\frac{1}{ab}$**

Where R is triangular region bounded by lines $x=0, y=0$ & $ax+by=1$

⑤ Evaluate $\iint_R (x^2+y^2)\, dx\, dy$ **Ans : 7/6**

Where R is region of tringle whose vertices are (0,0), (1,1) & (1,2).

⑥ Evaluate $\iint_R \frac{xy}{\sqrt{1-y^2}}\, dx\, dy$ **Ans : 1/6**

Where R is region of circle $x^2+y^2=1$ in first quadrant.

⑦ Evaluate $\iint_R y\, dx\, dy$, where R is region bounded by two parabolas $y^2=4x$ & $x^2=4y$. **Ans : $\frac{48}{5}$**

⑧ Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} y^2\, dx\, dy\, dx$

By changing the order of integration **Ans $\pi/16$**

⑨ By changing the order of integration, evaluate $\int_0^\infty \int_0^x x\, e^{-x^2/y}\, dy\, dx$ **Ans : $\frac{1}{2}$**

⑩ Evaluate $\int_0^a \int_y^a \frac{x}{x^2+y^2}\, dx\, dy$, By changing the order of integration **Ans : $\frac{\pi a}{4}$**

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(11) Evaluate $\int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} dy\,dx$, By changing the order of integration

Ans: $\frac{16a^2}{3}$

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(12) Evaluate $\iint_R (x^2+y)\,dx\,dy$, By changing into polar Form

where R is region of circle $x^2+y^2=1$

Ans: $\pi/4$

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(13) Evaluate $\int_0^a \int_y^a \frac{x^2}{(x^2+y^2)^{3/2}}\,dx\,dy$,

By changing in to polar form

Ans $= a/\sqrt{2}$

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(14) Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)}\,dy\,dx$, By changing into polar form

Ans: $\pi/4$

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(15) Evaluate $\int_0^2 \int_1^2 \int_0^1 x^2 y\,dz\,dy\,dx$

Ans $= 4$

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(16) Evaluate $\int_0^1 \int_{y^2}^1 \int_0^{1-x} x\,dz\,dx\,dy$

Ans $= 4/35$

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(17) Evaluate : $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz\,dz\,dy\,dx$

Ans: $\frac{1}{48}$

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(18) Evaluate : $\int_0^3 \int_{1/x}^1 \int_0^{\sqrt{xy}} xyz\,dz\,dy\,dx$

Ans: $\frac{13}{8} - \frac{\log 3}{6}$

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(19) Evaluate $\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{(x+y+z)}\,dz\,dy\,dx$

Ans: $\frac{8}{3}\log 2 - \frac{19}{9}$

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(20) Determine an area of plate in the form of a quadrant of ellipse

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Ans: $\frac{\pi ab}{4}$

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(21) Determine a smaller area bounded by ellipse $4x^2+9y^2=36$ and a straight line $2x+3y=6$

Ans: $\frac{3}{2}(\pi-2)$

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Determine an area lying betn two parabolas $y^2=4ax$ & $x^2=4ay$

Ans : $\frac{16}{3}a^3$

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(23) Determine a mass of circular plate $x^2+y^2=a^2$ lying in first quadrant of xy-plan. also find a centre of gravity. if density of a plate is const.

Ans : mass $= \frac{ka^2\pi}{4}$

$\left(\bar{x} = \frac{4a}{3\pi}, \bar{y} = \frac{4a}{3\pi}\right)$

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(24) Determine a mass of a lamina bounded by curves $y=x^2$ & $x=y^2$ if density at any point is $k(x^2+y^2)$

Ans : $\frac{6k}{35}$

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(25) Determine the co-ordinate of centre of gravity of the area bounded by curve $y^2=x$ & $x+y=2$ if density of area is constant (k)

Ans : mass $= 9/2$

$\bar{x} = 8/5$, $\bar{y} = -1/2$

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(26)

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